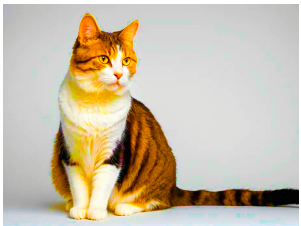
# Flash Chroma: joint tuning of four-stage LUTs

## From sparse color models to efficient image mapping



Input image



Four-stage output

**0.975 ms**

Fixed-chain mapping per image

**48 MiB**

Complete RGB8 lookup table

Complete project report  Evidence from retained fixed-chain and multi-LUT experiments.

# Research questions and report structure

### Task and representation

How can four color transforms be approximated?

How is integer stage behavior preserved?

### Algorithms and refinement

Sparse sampling and tetrahedral interpolation

Last-stage fitting and integer node search

### Evidence and scope

Accuracy, tails and runtime cost

Multi-LUT evaluation and demonstration

---

Core idea: **refine sparse models against the final chain output, then expand to a color lookup table.**

The delivered configuration has four fixed $33^3$ grids. Candidate exploration and subsequent studies are reported separately.

# Task definition

Given four ordered LUTs, **reduce per-image mapping time while controlling approximation error.**

**Input image**



**Reference output of the original chain**



**RGB8**
Integer inputs and stage outputs

**Pixelwise mapping**
Color-dependent; no spatial context

**Reuse across images**
One LUT chain, multiple images

Evaluation: final-output Euclidean RGB error, including its tail, and mapping time per image.

# Reference mapping and integer stage semantics

## The original four-stage chain defines the reference

$$z_0 = x, \qquad z_k = F_k(z_{k-1}), \quad k = 1, \ldots, 4, \qquad F(x) = z_4.$$

Each stage consumes and produces RGB8. The next stage receives the preceding integer output.

### Reference chain

Evaluate the original LUTs with their sampling and numeric-conversion semantics.

### Approximate chain

Reconstruct from sparse nodes by tetrahedral interpolation. Round and clip every stage to RGB8.

Both chains depend only on color. Interpolation, endpoints and rounding are part of the task semantics.

# Evaluation: mean and tail error

$$e(x) = \|S(x) - F(x)\|_2, \qquad R, G, B \in \{0, \ldots, 255\}.$$

| | |
|---|---|
| **Mean and SSE** | $\mu = N^{-1} \sum_x e(x)$ $\quad$ $\mathrm{SSE} = \sum_x e(x)^2$ |
| **p99 / max** | The 99th-percentile error and the maximum over all inputs. |
| **Worst-1% mean** | Average the largest $\lceil 0.01N \rceil$ errors. |
| **Image error** | Average pixels within each image, then weight images and chains equally. |

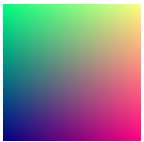
This is Euclidean RGB error, not CIE Lab $\Delta E$. A lower mean does not ensure improvement for every pixel or image.

# Method overview

Approximate four ordered color transforms with **low error and fast image mapping.**

## Preprocessing: construct a lookup table from color mappings

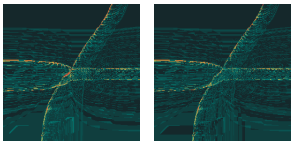
### Sparse candidates



$17^3$ / $33^3$

Fixed grids
Adaptive grids

### Chain-level refinement



Same color slice: sampled and refined errors

### Full-domain expansion

$256^3 \times 3$

RGB8 colors → RGB8 outputs

**48 MiB** direct table

## Image processing: one lookup per pixel



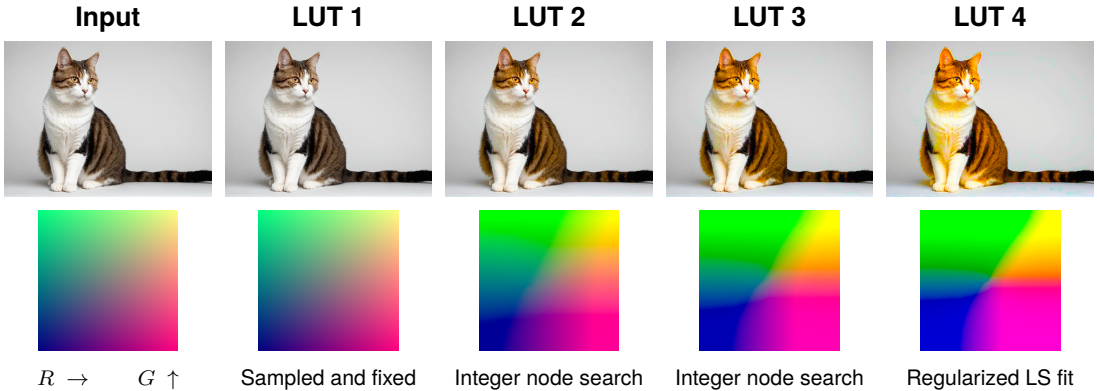
$$O[p] \leftarrow T[I[p]]$$

Depends only on pixel color; reuse the table across images.



# Layer-by-layer color visualization

Cumulative final1 outputs after stages 1 through $k$. Bottom: original input plane $B = 128$; horizontal $R$, vertical $G$, both 0–255.




**Each stage: a $33^3$ grid, tetrahedral interpolation, RGB8 rounding.** Fit stage 4, then search stages 3 and 2.

# Sparse grids: capacity and node placement

## Fixed sampling

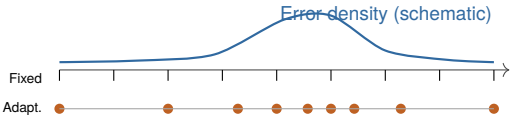
Fixed axis nodes. All four stages in the delivered model use $33^3$.

$$g = (0, 8, 16, \ldots, 248, 255).$$

The last cell spans 7 units, preserving the framework endpoint rule.

## Error-adaptive sampling

Allocate more axis nodes to regions with concentrated approximation error.




Candidate set: **fixed / adaptive** $\times$ $17^3/33^3$.

Adaptive grids are an exploratory route and do not replace the frozen configuration.

# Adaptive sampling pseudocode

```
function AdaptiveGrid(F, m, X_pilot)
    S ← UniformSample(F, m)
    e(x) ← ‖ Tetra(S, x) − F(x) ‖_2
    for a ∈ {R, G, B} do
        ρ_a ← Regularize(E[e(x) | x_a])
        g_a ← InverseCDF(ρ_a, m)
        g_a ← EndpointsAndSpacing(g_a)
    end for
    return Sample(F, (g_R, g_G, g_B))
```

### Sampling signal

Estimate error on $32^3$ probe colors and aggregate along each axis.

### Node constraints

Preserve endpoints and minimum spacing to avoid coincident nodes.

Regularized error density and inverse-CDF placement define nodes. Evaluate the resulting candidate at chain level.

# Select chains by final-output error

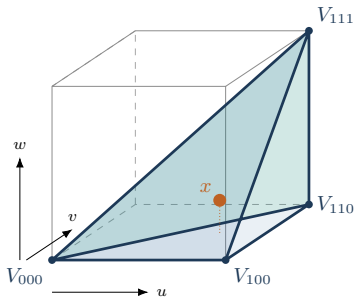
Four candidates per stage give $4^4 = 256$ four-stage combinations.

$$J(S) = \frac{1}{|X_{\mathrm{cal}}|} \sum_{x \in X_{\mathrm{cal}}} \|S(x) - F(x)\|_2$$

```
function SelectChain(F_1, ..., F_4, X_cal)
    C_k ← FourCandidates(F_k),  k = 1:4
    S* ← ∅;    J* ← +∞
    for S ∈ C_1 × ··· × C_4 do
        j ← MeanFinalError(S, F, X_cal)
        if (j, |S|) <_lex (J*, |S*|) then
            S* ← S;    J* ← j
        end if
    end for
    return S*
```

$X_{\mathrm{cal}}$ contains 32,768 random RGB8 colors. Equal-error ties favor smaller models. Sum of stage errors is not the chain objective.

# Geometry of tetrahedral interpolation




### Six coordinate orders per cell

For $u \geq v \geq w$, choose vertices $000, 100, 110, 111$.

$$\alpha = (1-u,\ u-v,\ v-w,\ w)$$

$$\alpha_j \geq 0, \qquad \sum_j \alpha_j = 1$$

Orange marks the query color. Four blue vertices define its tetrahedron.

$$\widehat{f}(x) = Q_8((1-u)V_{000} + (u-v)V_{100} + (v-w)V_{110} + wV_{111})$$

# A unified tetrahedral interpolation routine

1 **function** TETRA($S, x$)
2 $(i, f) \leftarrow$ LOCATECELL($S.\text{axes}, x$)
3 $\pi \leftarrow$ STABLESORTDESCENDING($f$)
4 $(u, v, w) \leftarrow (f_{\pi_1}, f_{\pi_2}, f_{\pi_3})$
5 $p_0 \leftarrow i; \quad p_1 \leftarrow p_0 + e_{\pi_1}$
6 $p_2 \leftarrow p_1 + e_{\pi_2}; \quad p_3 \leftarrow i + (1, 1, 1)$
7 $\alpha \leftarrow (1 - u, u - v, v - w, w)$
8 **return** $Q_8(\sum_{j=0}^{3} \alpha_j S[p_j])$

---

Use Q16 local coordinates. Integer weights satisfy $\sum_j w_j = 65536$.

$$y_c = \left\lfloor \frac{32768 + \sum_j w_j V_{j,c}}{65536} \right\rfloor .$$

$e_a$ is an axis unit vector. $Q_8$ rounds and clips to RGB8 at every stage.

# Last-stage fitting compensates chain error

## Fix the first three stages and fit stage 4 to reference outputs

$$z(x) = S_3(S_2(S_1(x))), \qquad y(x) = F(x).$$

### Linear surrogate

Build sparse matrix $A$ from the tetrahedral weights at $z(x)$, with at most four nonzeros per row.

### Anchor the initial nodes

Regularization restrains node movement and stabilizes sparsely covered nodes.

$$\min_{v_c} \|Av_c - y_c\|_2^2 + \sum_i \lambda_i (v_{i,c} - v_{i,c}^0)^2$$

This surrogate precedes last-stage rounding. Acceptance uses the actual RGB8 chain. The method changes stage 4 rather than adding a correction stage.

# Last-stage fitting pseudocode

```
function FitLast(S, F)
    (A, y) ← BuildFullDomainSystem(S_{1:3}, F)
    λ_i ← 10^{-5}((A^⊤ A)_{ii} + 1)
    for c ∈ {R, G, B} do
        v_c ← JacobiPCG(A, y_c, λ, v_c^0)
        v_c ← RoundAndClip(v_c, 0, 255)
        v_c ← IntegerRefine(v_c, maxSweeps = 8)
    end for
    C ← S with C_4 ← v
    if μ(C) ≤ μ(S) ∧ SSE(C) ≤ SSE(S) then
        return C
    end if
    return S
```

PCG uses at most 160 iterations and a relative-residual target of $10^{-8}$. Fitting guards mean and SSE. Five guards apply to subsequent node search.

# Integer node search refines the final output

```
1  function RefineChain(S, F)
2      S ← FitLast(S, F)
3      a ← Metrics(S, F, RGB8)
4      q_cur ← a
5      for k ∈ [3, 2] do
6          for C ∈ IntegerNodeProposals(S, k) do
7              q ← Metrics(C, F, RGB8)
8              if Guards(q, a) ∧ Better(q, q_cur) then
9                  S ← C;   q_cur ← q
10             end if
11         end for
12     end for
13     return S
```

### Bounded search

At most 32 candidate nodes per stage and one sweep.

26 RGB directions with steps 1, 2 and 4.

Evaluate final outputs through the actual suffix.

Metrics abstracts incremental evaluation and full-domain verification. The candidate-node budget differs from the number of modified nodes.

# Fixed-anchor guards and current-state improvement

## Guards: compare to a fixed anchor

$$q_j(C) \le a_j, \quad j \in \{\mu, \mathrm{SSE}, p_{99}, \max, \mathrm{CVaR}_{1\%}\}.$$

Anchor $a$ contains post-fit chain metrics and stays fixed after accepted updates.

## Better: compare to the current state

$$\mu(C) < \mu(S_{\mathrm{cur}})$$

Prioritize the current mean. Use SSE to break ties within $10^{-12}$.

**Acceptance covers all $256^3 = 16,777,216$ RGB8 inputs.**

These are project-specific multi-metric selection rules, not an unpublished official score. They do not impose pointwise monotonicity.

# Full-domain expansion preserves the refined chain

```
1  function Expand(S)
2      T ← RGB8Table(256^3)
3      for i = 0, ..., 256^3 − 1 do
4          x ← (i & 255, (i ≫ 8) & 255, i ≫ 16)
5          z ← x
6          for k = 1, ..., 4 do
7              z ← Tetra(S_k, z)
8          end for
9          T[i] ← z
10     end for
11     return T
```

$$T[\mathrm{index}(x)] = S(x) \quad \forall x \in \mathrm{RGB8}.$$

Expansion adds no approximation error. It exactly represents the optimized sparse chain; the error versus the original reference is determined by $S$.

# Image mapping by direct color indexing

```
1 function MapImage(I, T)
2     O ← Copy(I)
3     for p ∈ Pixels(I) do
4         (r, g, b) ← I[p]
5         i ← r + 256g + 65536b
6         O[p] ← T[i]
7     end for
8     return O
```

### Index space

$$0 \le i < 2^{24}$$

Each $T[i]$ is an RGB triplet.

### Runtime conditions

Measured with one AVX2 thread. pair32 describes pixel-block organization.

The measured interval includes output copying, indexing and writes. Pseudocode gives semantics; the kernel organizes vector instructions.

# Time and space costs

## 48 MiB

RGB24 lookup table

$256^3 \times 3 = 50,331,648$ B

Trade storage for one lookup per pixel.

## 448,252 B

Sparse model accounting

Four $33^3$ grids plus model metadata

Runtime arena: 50,919,295 B.

## 21.17 s

One preparation observation

Fit 8.27 s; search 12.64 s

Expansion: approximately 0.265 s.

---

$$T_{\mathrm{total}}(n) = T_{\mathrm{prepare}} + \sum_{j=1}^{n} T_{\mathrm{map}}(I_j).$$

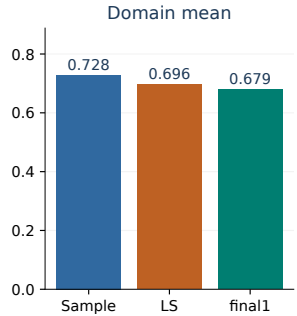
Report preparation and mapping separately. Accounting excludes source LUTs, reference tables and fitting temporaries; it is not peak process RSS.

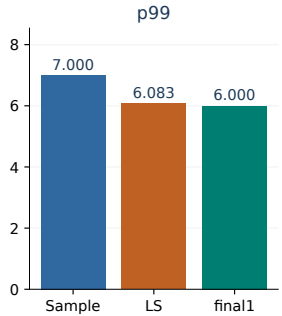
# Experimental design and data scope

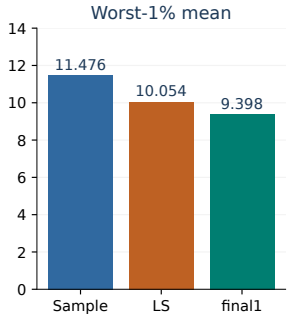
| | |
|---|---|
| **Fixed-chain accuracy** | The full RGB8 domain, 100 images, and four original images reported separately. |
| **Fixed-chain timing** | i9-14900K, Ubuntu, GCC 13.3, -O2/C++17, one P-core, AVX2. |
| **Multi-LUT batch** | 12 LUTs, eight four-stage chains, 24 images per chain; one warmup and 15 timed repetitions per image. |
| **Parameters and selection** | Fitting and node search use full-domain LUT error. The 100 images informed final historical selection; batch images reuse prior data. |

This report summarizes retained experiments. Benchmarks were not rerun while preparing these slides.

# Fixed-chain accuracy: sampling, fitting and search




**0.679297**
Domain mean

**1.481250**
100-image mean

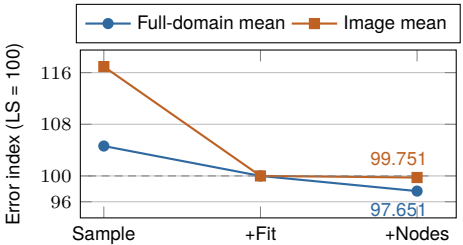
**50.209561**
Domain maximum

Versus direct $33^3$ sampling, final1 reduces domain mean by about 6.67% and image mean by about 14.69%. The original four-image mean is 1.481735.

# Component ablation and exploratory alternatives

Fixed four-stage chain, $33^3$ nodes/stage, 100 images; LS = last-stage least-squares fit. Lower is better.

**(a) Incremental components**




**(b) Independent alternatives from LS**

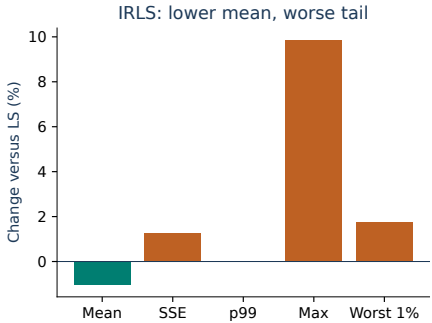


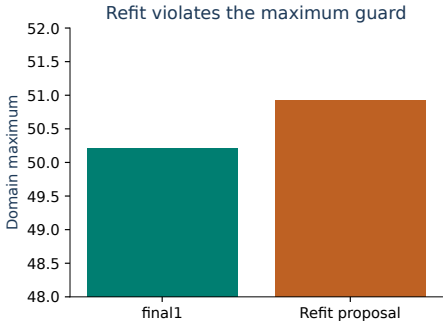

**Node search: further tail reduction**
Image mean $-0.249\%$; full-domain worst-1% mean $-6.53\%$.

**Lower mean does not ensure acceptance**
Refit violates the max guard; IRLS fails tail guards.

Right: independent alternatives with differing stage orders and constraints, not isolated causal effects. Refit is rejected; the model remains unchanged.

# A lower mean can violate tail constraints




## IRLS

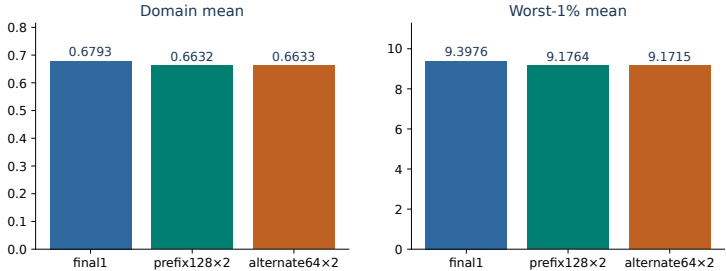
The image mean reaches 1.451506, but domain max rises to 55.154329 and the worst-1% mean also increases.

## Refit after node search

Domain mean falls to 0.675386, but max rises to 50.931326. The proposal is rejected.

IRLS is compared with LS; refitting is compared with final1. The experiments use different baselines.

# Subsequent experiments with larger search budgets




| Configuration | Domain mean | Image mean | Worst 1% |
|---|---|---|---|
| final1 | 0.679297 | 1.481250 | 9.397558 |
| prefix128×2 | 0.663217 | 1.478759 | 9.176440 |
| alternate64×2 | 0.663275 | 1.478341 | 9.171488 |

Both research candidates pass five guards relative to final1 and retain max 50.209561. They do not replace the frozen delivery. Preparation cost and multi-chain behavior require separate evaluation.

# Grid and neural-residual exploration

### Grid variants

**28**

Fitted candidates

Fixed, mixed-capacity and coordinate variants are checked over the full RGB8 domain.

**0/28 pass all five guards against final1.**

### Residual MLP exported to stage 4

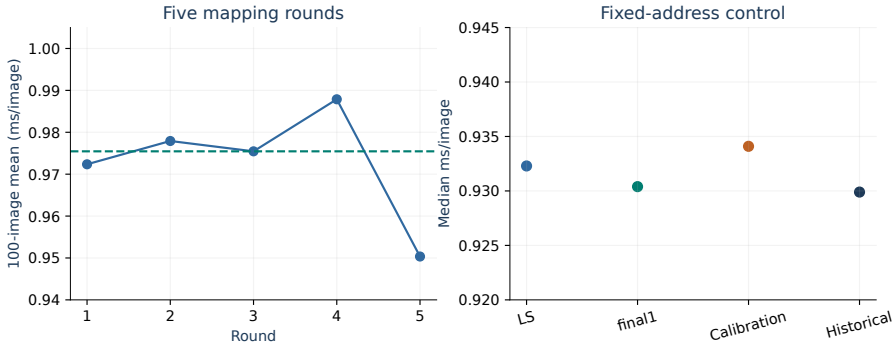
Width 32, two seeds, 4,000 steps each. Checkpoint selection uses color samples only.

One selects step 500: mean 0.679068, but SSE and worst-1% guards fail.

The other selects step 0 and retains the baseline.

Negative results concern these architectures, budgets and data. The network generates nodes and does not run during image mapping.

# Mapping time and fixed-address controls




**0.975453 ms/image** Median of five 100-image round means.

$1024 \times 768$ images. Includes output copying; excludes loading, preparation, reference evaluation and scoring. No extra apply warmup.

With the same input and table address, model medians differ by about 0.5%, below the observed A/A variation. This does not establish a stable speed advantage.

# Representation study: RGB24 versus RGB32

## 48 MiB

RGB24

Three output bytes per color.

**0.996792 ms/image**
Mean across ABBA samples

## 64 MiB

RGB32

Four bytes per color to study aligned storage.

**0.999423 ms/image**
Mean across ABBA samples

---

The 95% interval for the paired geometric RGB32/RGB24 ratio is **[0.99915, 1.00615]**.

The evidence supports retaining smaller RGB24 storage and does not show a stable RGB32 benefit.

Same process and frozen model, 100 images per sample. All 28 samples are retained and outputs match. This protocol differs from the five-round delivery timing.

# Design of the multi-LUT evaluation

**12**

Distinct LUTs

Generate dense RGB8 references under explicit input-range, interpolation and quantization policies.

**8**

Four-stage chains

Fit each chain separately and evaluate all $256^3$ colors.

**24**

Images per chain

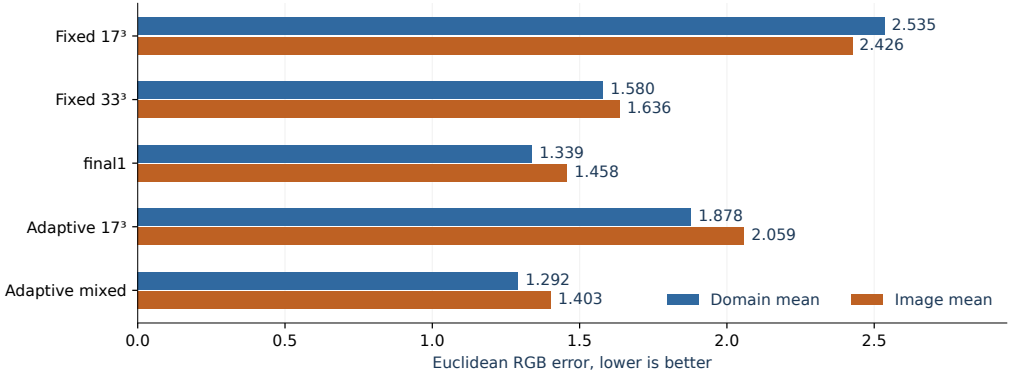
Reuse 20 earlier images and four original images, giving 192 chain–image pairs.

---

Five routes: fixed $17^3$, fixed $33^3$, final1, adaptive $17^3$ and adaptive mixed.

These images are not an independent holdout. Results describe the tested inputs rather than a guarantee for arbitrary LUTs.

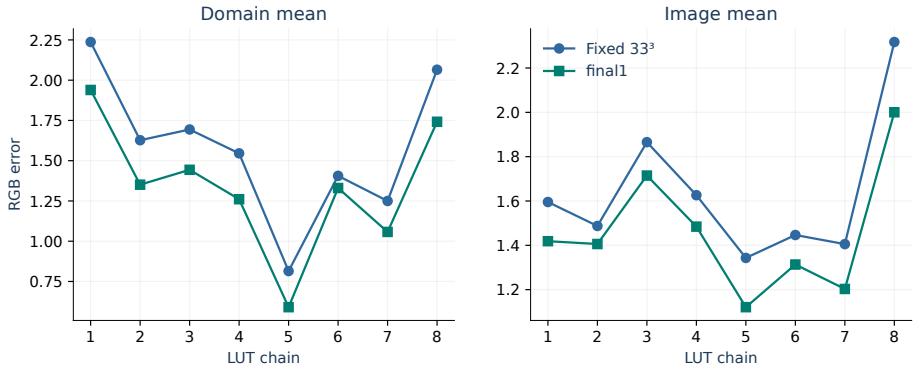
# Comparing domain and image means across routes




Equal weight across eight chains. Relative to fixed $33^3$, final1 reduces domain mean by 15.22% and image mean by 10.89%.

Adaptive mixed has a lower mean; its tail behavior and preparation cost are assessed separately.

# Mean error changes for each chain

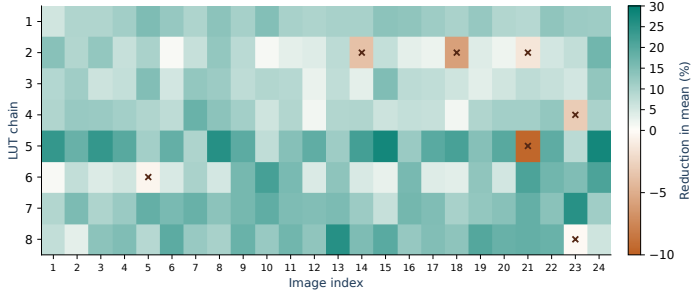



**8 / 8**

Chains with lower image mean

Gains vary by chain. LUT nonlinearity, quantization and composition order all affect error.

# Per-image results include gains and regressions

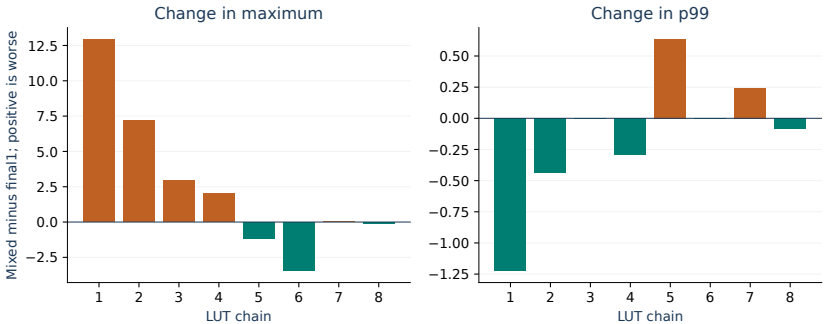



$$\text{gain} = 100\frac{\mu_{\text{sample33}} - \mu_{\text{final1}}}{\mu_{\text{sample33}}}.$$

**185 / 192** pairs improve; **7** pairs worsen and are marked by crosses.

Positive means lower error; negative means higher error. Reused images prevent treating pairs as independent random samples for significance claims.

# Adaptive mixed: mean versus tail tradeoffs




**1.402711**

Mixed image mean

**5 / 8**

Chains with higher max

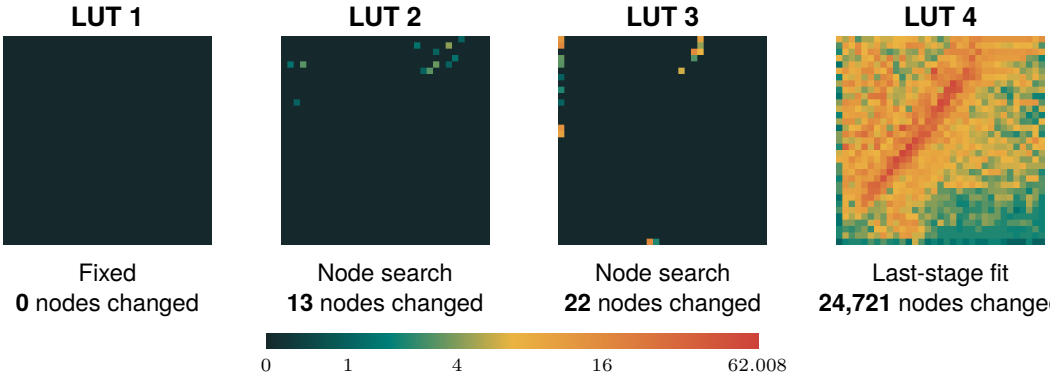
**2 / 8**

Chains with higher p99

The final1 image mean is 1.457651. Bars show per-chain differences. A lower aggregate mean does not replace chain-specific tail guards.

# Spatial distribution of node updates

Actual node updates: at each $(R, G)$ location, take the maximum RGB update norm over $B$; all panels share one scale.

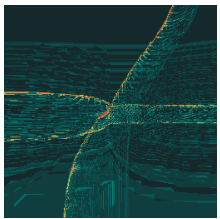



Node update $d = \|v_{\text{final1}} - v_{\text{sample}}\|_2$; color uses a $\log(1 + d)$ scale. Horizontal: $R$ nodes; vertical: $G$ nodes. Compare final1 nodes against initial $33^3$ sampling. These are node changes, not final-output image errors.

# Error changes on a fixed color slice

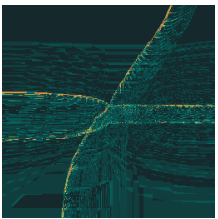
Fixed $B = 128$, horizontal $R$, vertical $G$. The first three maps share a 0–16 scale and clip higher values.

**Sample**



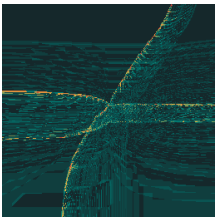
Mean 0.950257

**Last-stage fit**



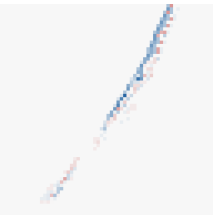
Mean 0.882954

**final1**



Mean 0.877152

**Local difference**



final1 − LS

First three: RGB error




Difference: final1 − fit

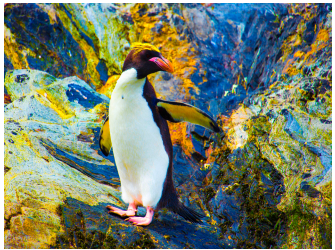



Difference zoom: $R = 152$–$215$, $G = 192$–$255$. This slice has 130 improved colors, 68 worse colors and 65,338 unchanged.

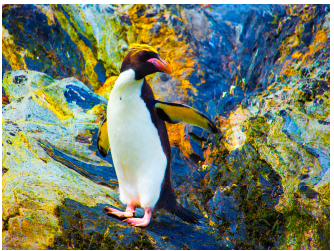
# Worst image among the 100

The image with the largest mean error among the 100. The error map is aligned pixel by pixel.

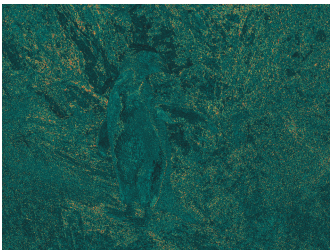
**Reference**



**final1**



**RGB error**




**2.386**

Image mean

**13.038**

Image p99

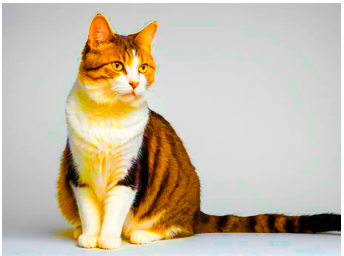
**48.021**

Image maximum

**Fitting and node search reduce aggregate error, without guaranteeing improvement for every color.**

# Cat image: mean and tail can move in opposite directions

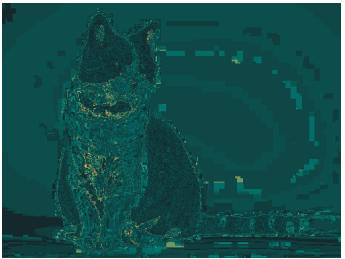
**Reference**



**final1**



**RGB error**



| | Image mean | p99 | max |
|---|---|---|---|
| Direct sampling | 0.708207 | 6.708204 | 36.469165 |
| final1 | 1.592951 | 5.385165 | 33.120990 |

The mean rises while p99 and max fall. **Domain-level guards do not ensure a lower mean for each image.**

# Interactive demonstration: image and video LUTs

## Images







Before/after, RGB error heatmap and metrics

## Video



**150** frames $960 \times 540$

Upload, framewise LUT mapping, counts, elapsed time and export.

## Open interactive demonstration

Derived presets demonstrate interaction. Accuracy plots use the frozen original. Video time includes decoding, mapping and encoding.

# Scope and limitations

## Error and data

Euclidean RGB error is not a direct perceptual-quality measure.

Images informed historical comparisons. The extension reuses images, so new held-out images and LUTs are needed.

Bounded node search gives empirical results, without proving a global four-stage optimum.

## Cost and use cases

A 48 MiB table and second-scale preparation favor repeated use of one chain. Frequent LUT changes require total-cost evaluation.

One CPU's mapping result does not establish performance on other CPUs, phones or full video pipelines.

Further evaluation should jointly consider mean, tails, preparation, memory and stability across settings.

# Conclusions: model accuracy and execution representation

### Optimize sparse LUTs against the final chain output

Regularized last-stage fitting and intermediate integer search reduce error under multiple aggregate guards.

### Expand the color domain to reduce per-image computation

The frozen model uses one lookup per pixel and records 0.975453 ms/image under the fixed setup, including output copying.

### Report gains, regressions and rejected candidates

All eight chain image means improve; 185/192 pairs improve. Lower-mean alternatives still require tail and construction-cost evaluation.

The results support a reusable accuracy–time tradeoff, not optimality for arbitrary LUTs.

# Appendix: algorithm parameters

| | |
|---|---|
| **Grids and endpoints** | Four $33^3$ grids; each axis is $0, 8, 16, \ldots, 248, 255$. |
| **Interpolation** | Stable coordinate ordering, Q16 weights, four tetrahedral vertices, RGB8 rounding at each stage. |
| **Last-stage solve** | Per-channel Jacobi-PCG, at most 160 iterations, $r_{\mathrm{tol}} = 10^{-8}$; at most eight integer sweeps. |
| **Regularization** | $\lambda_i = 10^{-5}(H_{ii} + 1), \quad H = A^{\top}A.$ |
| **Integer node updates** | Stages 3 then 2; 32 candidate nodes/stage, one sweep; 26 directions, steps 1, 2, 4. |
| **Acceptance and runtime** | Full RGB8 domain, five fixed-anchor guards; DirectRGB24, one AVX2 thread. |

# Appendix: complete ablation results

| Configuration | Domain mean | Image mean | Domain max | Worst 1% |
|---|---|---|---|---|
| $33^3$ sampling | 0.727840 | 1.736378 | 66.037868 | 11.476347 |
| Last-stage fit (LS) | 0.695639 | 1.484947 | 50.209561 | 10.053581 |
| LS + node search (final1) | 0.679297 | 1.481250 | 50.209561 | 9.397558 |
| Refit: retained model | 0.679297 | 1.481250 | 50.209561 | 9.397558 |
| Pointwise-guard search | 0.695501 | 1.484945 | 50.209561 | 10.040070 |
| Regional-guard search | 0.695270 | 1.483276 | 50.209561 | 10.029045 |
| Shared-coordinate calibration | 0.660909 | 1.489587 | 50.209561 | 9.335890 |
| Per-output-channel calibration | 0.664432 | 1.481943 | 50.209561 | 9.415169 |
| IRLS (constraints violated) | 0.688486 | 1.451506 | 55.154329 | 10.228190 |

Raw refit proposal: domain mean 0.675386, max 50.931326; rejected by the max guard. No 100-image measurement.

Independent alternatives: pointwise/regional search uses stages 4, 3, 2, 1; calibration alternates updates in stages 4, 3, 2; IRLS refits the last stage with a fixed prefix.

# Appendix: aggregate batch results

| Route | Domain mean | Image mean | Worst image | Largest max |
|---|---|---|---|---|
| Fixed $17^3$ | 2.534900 | 2.425578 | 5.493720 | 101.439637 |
| Fixed $33^3$ | 1.579881 | 1.635849 | 2.903687 | 74.067537 |
| final1 | 1.339346 | 1.457651 | 2.664704 | 57.297469 |
| Adaptive $17^3$ | 1.878148 | 2.058791 | 4.302554 | 96.529788 |
| Adaptive mixed | 1.291718 | 1.402711 | 2.465986 | 70.242437 |

Means weight chains equally. Worst-image mean and largest domain max take the maximum over all chains.

Integrity records contain 40 full-domain checks, 960 image results and 14,400 direct timing samples. Recorded direct/sparse output mismatch counts are zero.

# Appendix: collisions and a fixed-prefix lower bound

## Distinct inputs can collide after the quantized prefix

If $P(x_1) = P(x_2)$, a deterministic last stage must give both inputs the same output.

$$P = S_3 \circ S_2 \circ S_1, \qquad \min_{v_z \in \mathrm{RGB8}} \sum_z \sum_{x:P(x)=z} \|v_z - F(x)\|_2.$$

**1,127,108**

Reachable prefix colors

214,344 groups contain multiple target outputs.

**0.34089797**

Relaxed optimum mean

$[0.340897968287,\ 0.340897970095]$

The evidence fixes the LS baseline's first three stages and allows an independent RGB8 output per reachable color. It does not apply after changing the prefix and is not a global-optimality certificate for four $33^3$ stages.

# Appendix: statistical definitions and evidence

Sort $N$ errors as $e_{(0)}, \ldots, e_{(N-1)}$.

$$j = \lfloor 0.99(N-1) \rfloor, \quad \alpha = 0.99(N-1) - j, \quad p_{99} = (1-\alpha)e_{(j)} + \alpha e_{(j+1)}.$$

$$K = \lceil 0.01N \rceil, \qquad \mathrm{CVaR}_{1\%} = \frac{1}{K} \sum_{i=N-K}^{N-1} e_{(i)}.$$

| | |
|---|---|
| **Numerical evidence** | Configuration, raw timings, full-domain errors, image results and node updates are retained separately and supply the figures. |
| **Mean interpretation** | Domain means weight colors uniformly. Image means reflect observed color frequencies, giving different objectives. |
| **Scope** | Fixed-chain results, multi-chain results, derived demo presets and relaxed bounds use distinct baselines. |